
Software Requirements Specification

for

Distributed Information System for Banking Transactions

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Dobre Marina-Iulia			
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the comprehensive functional and non-functional requirements for the **Distributed Information System for Banking Transactions**. The purpose of this system is to provide clients with a secure and efficient platform to manage their accounts and conduct online financial transactions.

1.2 Document Conventions

This document follows the standard IEEE formatting for software requirements. The following typographical conventions and standards are applied throughout the SRS:

- **Requirement Tagging:** Every individual requirement is uniquely identified with a tag in the format **REQ-[Number]** (e.g., REQ-1, REQ-2) to ensure they are verifiable and easy to track.
- **Priority Definition:** Every functional requirement statement carries its own priority rating (High, Medium, or Low) within its respective feature section. Priorities are not inherited from higher-level features but are assigned based on the requirement's necessity for core banking operations.
- **Terminology:** Standard banking terms (e.g., IBAN, Debit, Credit) are used as defined in the distributed system context.

1.3 Intended Audience and Reading Suggestions

This document is for anyone helping to build or test the banking system. It explains how the website, the security tools, and the data servers work together to let customers manage their money.

- **Developers:** Use this to see how the screens should connect to the database and how the money-transfer logic works.
- **Project Managers:** Read the first two sections for a quick summary of what the system does and who it is for.
- **Testers:** Use the list of 60 features in Section 4 to make sure every button and security check works exactly as planned.
- **Documentation Writers:** Use Sections 2 and 3 to write the help guides and manuals for the customers.

How to read this document:

- **Start with the basics:** Read Sections 1 and 2 to understand the "big picture" of the banking app.
- **Check the details:** If you are building the app, go straight to Section 4 for the 60 specific rules and features.
- **Security focus:** Everyone should look at Section 5 to understand how we keep client passwords and money safe.

1.4 Product Scope

The Distributed Information System for Banking Transactions is a multi-tier software platform designed to automate and simplify online financial activities for bank clients. Its primary purpose is to allow users to securely manage their accounts and perform transactions from any location with internet access.

1.5 References

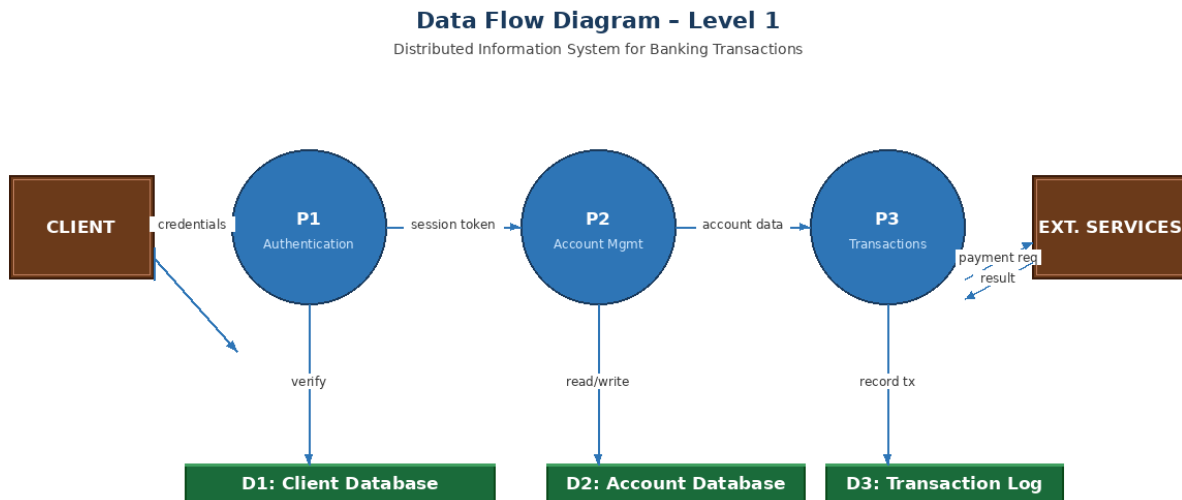
- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- Karl E. Wiegers, Software Requirements (Microsoft Press, 1999).
- OWASP Top 10 Web Application Security Risks (2021).
- RFC 2818 – HTTP Over TLS.
- GDPR (EU) 2016/679 – General Data Protection Regulation.
- [ING Home Bank - serviciul de internet banking gratuit](#)

2. Overall Description

2.1 Product Perspective

The Distributed Information System for Banking Transactions is a self-contained software product designed to provide comprehensive digital financial services. The system replaces traditional manual banking methods with an automated platform that allows clients to check their accounts and perform transactions entirely online. The context of this product is a multi-tier distributed architecture where the software is divided into specialized components to ensure security and efficient data management.

The system follows a distributed architecture where the client interface communicates with the web server through HTTPS requests. The web server forwards requests to the application server, which processes the logic and interacts with the database server for storing and retrieving data, as seen in the diagram below:



2.2 Product Functions

- **Secure Authentication and Access:** The system provides a secure gateway for clients to log in using a unique identifier and a password, ensuring that only authorized individuals can access private financial data;
- **Multi-Currency Account Management:** Clients can view and manage multiple accounts simultaneously, including specific account types such as debit, credit, and loan accounts in various currencies including EUR, USD, and RON;
- **Financial Transaction Processing:** The system allows for the execution of funds transfers between a user's own accounts or to external beneficiaries by communicating directly with the application server;
- **Utility and Service Payments:** Users can perform direct payments to specific service providers, such as Electrica or various telephony companies, by entering the required subscriber information;
- **Information Visualization:** The application retrieves real-time data from the database server to display account balances, transaction histories, and specific bank information to the user;
- **Profile Administration:** Clients have the ability to manage their personal profile settings, which includes up dating their residential address and changing their security credentials for account protection.

2.3 User Classes and Characteristics

The system is built primarily for the Bank Client, who is the most important person to satisfy. This user needs a simple and secure way to handle money without going to a physical bank.

Characteristics of the Bank Client:

- **Access:** They use a pre-set identifier and password to log in.
- **Technical Skill:** They only need basic experience using a web browser or a simple mobile app.
- **Frequency of Use:** They may log in daily to check balances or monthly to pay utility bills.
- **Tasks:** They manage various accounts in **EUR, USD, and RON**, including debit, credit, and loan accounts.
- **Responsibility:** They are responsible for keeping their password safe and logging out when finished.

2.4 Operating Environment

The banking system is a distributed application that runs across several different environments to ensure data is handled safely and efficiently. The software operates within the following setup:

- **Client Side:** Users access the system through a standard Web browser.
- **Web Server:** This layer handles the incoming traffic and hosts the user interface that clients interact with.

- **Application Server:** This is where the core logic lives, managing all the details for clients, accounts, and financial transactions.
- **Database Server:** A dedicated server used to store all persistent information about clients, their various accounts, and the bank itself.

2.5 Design and Implementation Constraints

There are several rules and limitations that the development team must follow when building this system:

- **Access Limitation:** Users are strictly forbidden from accessing the service through any means other than the provided user interface.
- **Authentication Logic:** The system must only provide banking facilities after a client successfully logs in using a specific identifier and password.
- **Pre-existing Data:** The system is designed with the assumption that client identifiers and passwords have already been generated.
- **Currency Support:** The database and application logic must support accounts in three specific currencies: EUR, USD, and RON.
- **Account Types:** The system must be able to categorize and manage different types of accounts, including debit, credit, and loan accounts.

2.6 User Documentation

The following documentation will be delivered together with the system: Online Help Section: A basic FAQ page embedded in the application, accessible from every screen.

2.7 Assumptions and Dependencies

The following assumptions apply to the development and operation of this system:

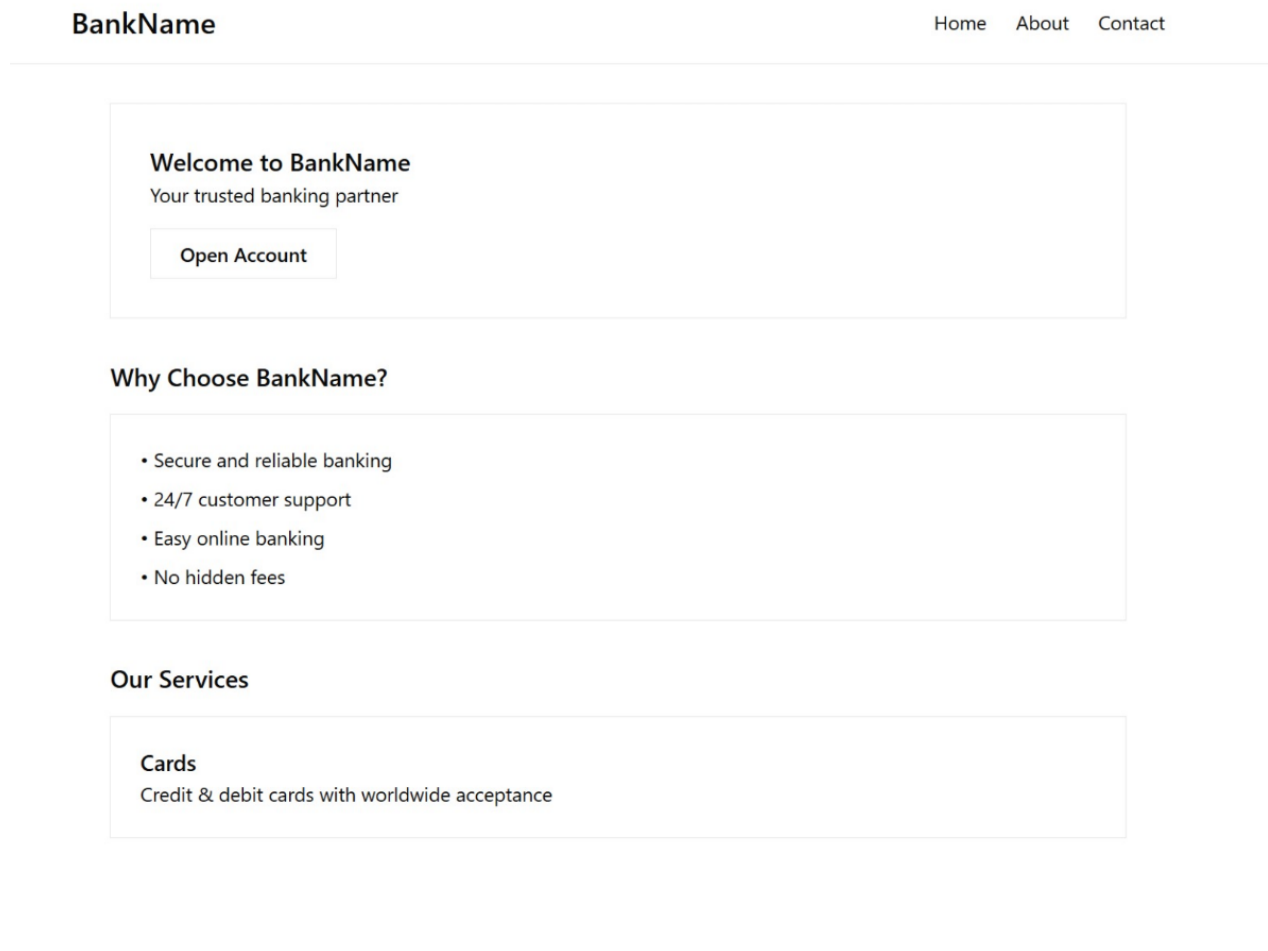
- Client identifiers and initial passwords are pre-generated by the bank and provided to clients separately.
- The hosting infrastructure (servers, network connectivity) is provided and maintained by the bank's IT department.
- All clients are assumed to have access to the internet and a compatible web browser.

3. External Interface Requirements

3.1 User Interfaces

HOME

The Home page serves as the public-facing entry point to the banking application. It displays a navigation bar with links to Home, About, and Contact sections, a welcome banner with an "Open Account" call-to-action button, a list of key services offered by the bank, and a footer with copyright information.



REGISTRATION

The Registration screen allows new clients to open a bank account. It includes input fields for First Name, Last Name, CNP (National ID), Email, Phone Number, Password, and Confirm Password, along with a Sign Up button and a link for existing users to log in.

BankNameHome About Contact

Sign Up

First Name

Last Name

CNP

Email

Phone Number

Password

Confirm Password

Sign Up

Already have an account? Log in

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AUTHENTICATION

The Authentication (Log In) screen provides fields for Client ID and Password, a Log In button, and a link to the registration page for new users.

BankNameHome About Contact

Log In

Client ID

Password

Log In

Don't have an account? [Sign up](#)

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PROFILE & PROFILE MANAGEMENT

The Profile page displays a Profile Overview panel showing the client's name, email, phone number, CNP, and address, along with an option to change the profile picture. Below, a My Cards panel lists all associated debit and credit cards with their last 4 digits, expiry date, and a Manage button. The Edit Profile Information form allows clients to update their personal details, and the Change Password form allows updating security credentials.

An additional panel shows My Accounts with the current balance for each RON, EUR, and USD account. Clients can also Request a New Card (debit or credit) and Open a New Account by selecting a currency.

BankNameHomeAboutContactLogged in as John Doe

Profile Overview

Profile Picture

John Doe

john.doe@email.com

+40 123 456 789

CNP: 1234567890123

123 Main Street, Bucharest

Change Profile Picture

My Cards

Debit Card ****1234 Expires: 12/2027	Active Manage
Credit Card ****5678 Expires: 08/2028	Active Manage
Debit Card ****9012 Expires: 03/2026	Active Manage

Request New Card

CARD TYPE

Select card type ▼

Request Card

My Accounts

RON Account ****1234	24,500.00 RON
EUR Account ****5678	3,200.00 EUR
USD Account ****9012	1,800.00 USD

Open New Account

CURRENCY

Select currency ▼

Open Account

Edit Profile Information
CHANGE FIRST NAME

CHANGE LAST NAME

CHANGE EMAIL

CHANGE PHONE NUMBER

CHANGE ADDRESS

Change Password
Current Password

New Password

Confirm New Password

TRANSFERS

The Transfers page contains a New Transfer form with dropdown selectors for the source and destination accounts, an amount field, and an optional description. Below it, the My Accounts summary shows balances, and a Recent Transfers section lists past transactions with date, account details, and amount.

BankNameHome About Contact Logged in as John Doe

New Transfer

FROM ACCOUNT

Select account

TO ACCOUNT

Select account

AMOUNT

Enter amount

DESCRIPTION (Optional)

Enter description

Transfer Funds

My Accounts

Checking Account ****1234	\$5,240.50
Savings Account ****5678	\$12,800.00
Personal Account ****9012	\$3,450.75

Recent Transfers

2026-03-04 From ****1234 to ****5678 Monthly savings	\$500.00
--	----------

BILL PAYMENTS

The Bill Payments page contains a Pay Bill form where the client selects the source account, the biller from a dropdown, enters the amount, and an optional reference number. A Saved Billers panel lists pre-saved service providers (e.g., Electric Company, Water Services, Internet Provider, Gas Company, Phone Service), each with a Remove button and an Add New Biller option. A Recent Payments section at the bottom displays past bill payments with date, provider, amount, and status.

All screens include a consistent navigation bar with "Logged in as [Name]" displayed after authentication. Error messages are shown inline near the relevant field. All monetary values are displayed with two decimal places and the appropriate currency symbol.

BankNameHome About Contact Logged in as John Doe

Pay Bill

FROM ACCOUNT

BILLER

AMOUNT

REFERENCE (Optional)

Saved Billers

Electric Company
123456789

Water Services
987654321

Internet Provider
456789123

Gas Company
789123456

Phone Service
321654987

Recent Payments		
2026-03-01	\$125.00	
Electric Company	Completed	
March 2026		
2026-02-28	\$65.00	
Internet Provider	Completed	
February 2026		
2026-02-25	\$45.00	
Water Services	Completed	
February 2026		
2026-02-20	\$80.00	
Gas Company	Completed	
February 2026		

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3.2 Hardware Interfaces

Does not apply.

3.3 Software Interfaces

The user accesses the application through a web browser or dedicated client application, which provides the graphical interface for performing operations such as authentication, viewing account information, and making transactions. The browser sends requests to the system and receives responses such as account details, transaction confirmations, or error messages.

The application server contains the main logic of the system. It processes user actions such as login, viewing account information, changing personal data, performing transfers, and making payments.

The application server interacts with a database server, which stores all persistent data used by the system. This includes client information, account details, account types (EUR, USD, RON), and transaction history.

A security module is used to verify user credentials during authentication and ensure that only authorized clients can access the banking services.

3.4 Communications Interfaces

The system communicates with users through the Internet using standard web protocols. Communication between the client interface and the web server is performed using HTTP/HTTPS requests.

The web browser or client application sends requests to the server when the user performs actions such as logging in, viewing account information, or making transactions. The server processes these requests and sends back responses containing the requested data or confirmation messages.

Secure communication is ensured through the use of HTTPS, which encrypts the data transmitted between the client and the system.

4. System Features

This section describes the main functional features of the distributed banking system. The system is organized around several core services that allow clients to interact with their bank accounts through an online interface. These features include client registration, authentication, account management, financial transfers, bill payments, and profile management. Each feature is described through a short description, the stimulus/response sequence that triggers the functionality, and the detailed functional requirements that define the system's behavior.

The following subsections present the main features supported by the system:

4.1 Client Registration (Account Opening)

4.1.1 Description and Priority

This feature allows new users to apply for a bank account and establishes their initial profile in the distributed database.

Priority: High

4.1.2 Stimulus/Response Sequences

Stimulus: User selects "Open New Account" from the UI.

Response: System displays a registration form requesting personal and initial account details.

Stimulus: User submits the completed form.

Response: System validates data, generates a unique Client ID, and stores information in the database server

4.1.3 Functional Requirements

REQ-1: The system shall provide a registration form for new clients.

REQ-2: The system shall require a full legal name for registration.

REQ-3: The system shall collect a unique national ID number (CNP).

REQ-4: The system shall require a valid email address for communication.

REQ-5: The system shall allow users to choose a primary account currency: EUR, USD, or RON.

REQ-6: The system shall automatically generate a unique Client ID.

REQ-7: The system shall require a password that meets length requirements.

REQ-8: The system shall verify that the user is at least 18 years old.

REQ-9: The system shall provide a "Terms and Conditions" checkbox.

REQ-10: The system shall save the new client profile to the database server.

4.2 Client Authentication

4.2.1 Description and Priority

Manages secure access to the distributed system using the generated identifier and password.

Priority: High

4.2.2 Stimulus/Response Sequences

Stimulus: User enters Client ID and password.

Response: The security module verifies credentials against the database server.

Response: System grants access to the main dashboard upon success.

4.2.3 Functional Requirements

REQ-11: The system shall provide a login interface for Client ID and password entry.

REQ-12: The system shall mask password characters during entry for visual security.

REQ-13: The system shall verify credentials against the distributed database server.

REQ-14: The system shall deny access and display a generic error message for incorrect credentials.

REQ-15: The system shall lock an account after three consecutive failed login attempts.

REQ-16: The system shall display the user's name on the dashboard upon successful login.

REQ-17: The system shall provide a manual "Log Out" button that revokes the current session.

REQ-18: The system shall generate a unique session token upon successful login.

REQ-19: The system shall show a "Logged in as [Name]" label in the header of all pages.

REQ-20: The system shall display a welcome message with the user's name on the dashboard upon successful login.

4.3 Account Management

4.3.1 Description and Priority

Provides tools for clients to monitor their balances and account types.

Priority: High

4.3.2 Stimulus/Response Sequences

Stimulus: User selects the account list.

Response: System displays all accounts and their current balances.

4.3.3 Functional Requirements

REQ-21: The system shall have debit and credit accounts for its clients.

REQ-22: Each client shall be able to create a debit or credit account.

REQ-23: The system shall list all EUR, USD, RON accounts owned by the client.

REQ-24: The system shall show the real-time balance for every account.

REQ-25: The system shall display the IBAN for each account.

REQ-26: The system shall allow users to open a secondary account in a different currency.

REQ-27: The system shall provide a summary of the total funds across all accounts.

REQ-28: The system shall display a "copy" button next to the IBAN for easy sharing.

REQ-29: The system shall show the account status (e.g., "Active").

REQ-30: The system shall allow clients be able to close an account of their preference.

4.4 Financial Transfers

4.4.1 Description and Priority

Enables the movement of funds to other bank accounts.

Priority: High

4.4.2 Stimulus/Response Sequences

Stimulus: User enters recipient IBAN and amount.

Response: System executes the transfer and updates balances.

4.4.3 Functional Requirements

REQ-31: The system shall provide a "New Transfer" form.

REQ-32: The system shall allow selecting a source account from a dropdown.

REQ-33: The system shall provide an input field for the recipient's name.

REQ-34: The system shall provide an input field for the destination IBAN.

REQ-35: The system shall validate that the transfer amount is greater than zero.

REQ-36: The system shall check for sufficient funds before processing.

REQ-37: The system shall provide a "Transfer Reference" text field.

REQ-38: The system shall display a confirmation summary before the final execution.

REQ-39: The system shall update the database in real-time after a successful transfer.

REQ-40: The system shall display a success message with a transaction ID.

4.5 Utility Bill Payments

4.5.1 Description and Priority

Allows clients to pay bills for integrated service providers.

Priority: Medium

4.5.2 Stimulus/Response Sequences

Stimulus: User selects a provider and enters details.

Response: System deducts funds and completes the payment.

4.5.3 Functional Requirements

REQ-41: The system shall list "Electrica" as a bill payment provider.

REQ-42: The system shall list "Telefonica" as a bill payment provider.

REQ-43: The system shall provide an "Other Services" (etc) category.

REQ-44: The system shall require a subscriber code for utility payments.

- REQ-45:** The system shall allow users to choose the payment currency.
- REQ-46:** The system shall validate the subscriber code format.
- REQ-47:** The system shall save frequently paid bills to a "Quick Pay" list.
- REQ-48:** The system shall allow users to view a history of bill payments.
- REQ-49:** The system shall display the provider's logo for easier identification.
- REQ-50:** The system shall generate a digital receipt for every bill paid.

4.6 Profile and History

4.6.1 Description and Priority

Manages personal settings and audit logs of transactions.

Priority: Medium

4.6.2 Stimulus/Response Sequences

Stimulus: User updates their address or views history.

Response: System updates the database or retrieves the requested log.

4.6.3 Functional Requirements

- REQ-51:** The system shall allow users to change their residential address.
- REQ-52:** The system shall allow users to change their password.
- REQ-53:** The system shall show a list of all recent transactions.
- REQ-54:** The system shall allow filtering transactions by date.
- REQ-55:** The system shall allow filtering transactions by currency (EUR, USD, RON).
- REQ-56:** The system shall provide a search bar to find transactions by recipient.
- REQ-57:** The system shall allow users to update their contact phone number.
- REQ-58:** The system shall display a profile summary with the Client ID.
- REQ-59:** The system shall provide a "Help/Support" link in the footer.
- REQ-60:** The system shall allow users to download a simple transactional statement in text format.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system should provide fast and reliable responses to user actions under normal operating conditions.

Most user operations, such as logging in, viewing account balances, or loading the dashboard, should be completed within 2–3 seconds. Database queries related to account information, transactions, and user profiles should return results within 3 seconds under typical usage conditions.

Financial operations such as transfers or bill payments should be processed and confirmed within 5 seconds, including database updates.

5.2 Safety Requirements

The system must ensure that financial operations are performed safely to prevent loss of funds or incorrect transactions.

Before executing any transfer or payment, the system must verify that the source account has sufficient funds. The system must also display a confirmation summary before finalizing any financial transaction to prevent accidental transfers.

User input must be validated to prevent invalid or incorrect data from being stored in the system. These safeguards help ensure that users do not accidentally cause financial loss or corruption of account information.

5.3 Security Requirements

The system must ensure the protection of user data and financial information at all times. Access to the application shall only be granted to authenticated users through the use of a unique Client ID and password.

Password characters must be masked during input to prevent unauthorized viewing.

The system must use secure communication protocols (HTTPS) to protect data transmitted between the client interface and the server. Sensitive information such as login credentials and personal data must never be transmitted in plain text.

The system must restrict access to personal account information so that users can only view and manage their own data.

5.4 Software Quality Attributes

The system should provide several quality characteristics that ensure reliability and ease of use for clients.

Usability:

The user interface should be simple and intuitive so that clients can easily perform operations such as viewing accounts, making transfers, or paying bills without requiring technical knowledge.

Reliability:

The system must operate consistently and ensure that all financial transactions are processed correctly. In case of errors or failures, the system must prevent incomplete transactions and maintain data integrity.

Availability:

The system should be available to users at most times, allowing clients to access their accounts and perform banking operations online without significant interruptions.

5.5 Business Rules

The system operates according to several rules that define how banking services are used and who can perform certain actions.

Only registered clients who have successfully completed the account opening process are allowed to access the system and perform banking operations. Each client is identified by a unique **Client ID** generated by the system.

Clients must be **at least 18 years old** to open a bank account. Each client may own one or more accounts, and these accounts can be in different currencies such as **EUR, USD, or RON**.

Financial transactions, such as transfers or bill payments, can only be performed if the source account contains sufficient funds. All transactions performed by a client must be recorded and stored in the transaction history.

Clients are allowed to modify certain personal information, such as their address or password, but they are not allowed to modify system-generated identifiers such as the **Client ID** or the **IBAN** of an account.

The system must ensure that users can only access and manage their own accounts and cannot view or modify the data of other clients.

6. Other Requirements

This section defines additional constraints that ensure the distributed system functions correctly across different regions and complies with data standards.

Data Retention: All transaction records must be retained for a minimum period as required by applicable financial regulations.

Localization: The system should display dates and currency values in formats consistent with Romanian standards (DD.MM.YYYY and the respective currency symbol).

Compliance: The system must comply with GDPR regulations regarding the storage and processing of personal client data. Clients must be able to request access to or deletion of their personal information.

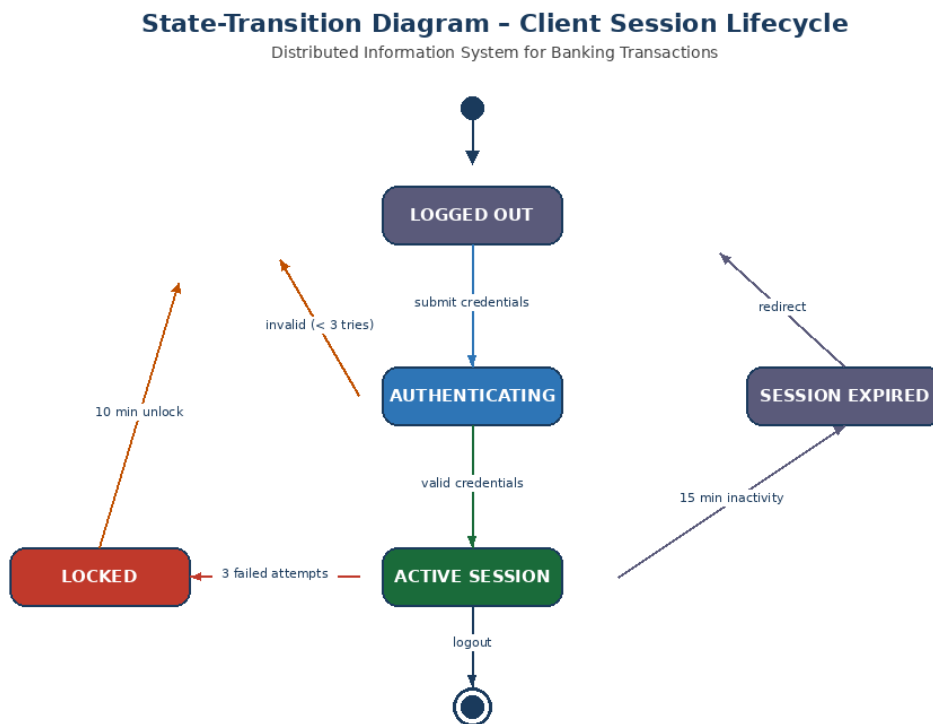
Appendix A: Glossary

The following terms are defined to help interpret the requirements for the banking system:

- **Application Server:** The central component that runs the business logic for managing clients and accounts.
- **Client ID:** A unique alphanumeric identifier assigned to each registered bank client, used as the login username.
- **CNP:** Cod Numeric Personal – the unique Romanian national identification number.
- **Database Server:** The storage unit used to keep permanent records of bank and client data.
- **Distributed System:** A software system where components are located on different servers (Web, App, DB) but work together as one.
- **HTTPS:** HyperText Transfer Protocol Secure – the encrypted version of HTTP used to protect data in transit.
- **IBAN:** International Bank Account Number, used to uniquely identify accounts for transfers.
- **Security Module:** A specific software layer that handles user authentication and data protection.
- **Session Token:** A temporary cryptographic token assigned to a user upon login that authorizes subsequent requests.
- **User Interface (UI):** The web or mobile screens that the client uses to interact with the bank.

Appendix B: Analysis Models

The state-transition diagram below describes the lifecycle of a client session. A client starts in the Logged Out state, moves to Authenticating when credentials are submitted, and transitions to Active Session on success. After 3 failed attempts the account becomes Locked for 10 minutes. The session ends either through manual logout or after 15 minutes of inactivity.



Appendix C: To Be Determined List

Does not apply.